

## RIO DE JANEIRO SANTA CRUZ, Brazil

WMO: 837410

Lat: 22.932S

Lon: 43.719W

Elev: 10

StdP: 14.69

Time Zone: -3.00 (SAB)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |      |      |      |      |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|------|------|------|------|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |      |      | 99%  |      |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR   | MCDB | DP   | HR   | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e)  | (f)  | (g)  | (h)  | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 7             | 56.8       | 58.8 | 50.3                          | 54.0 | 70.1 | 53.2 | 60.2 | 69.5 | 22.2                  | 74.9 | 20.2 | 74.5 | 4.3                   | 40   | 0.319 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 2             | 17.4                   | 97.1            | 76.8 | 95.1 | 76.4 | 93.1 | 76.4 | 80.9                | 91.3 | 79.8 | 89.6 | 78.9 | 88.2 | 10.0                 | 20   |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |       |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|-------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |       |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR    | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)   | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 77.4                            | 142.7 | 86.7 | 77.0 | 140.9 | 86.3 | 75.6 | 134.1 | 83.7 | 44.6          | 92.2 | 43.4 | 89.7 | 42.3 | 88.2 | 87.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |       |  |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|-------|--|
| 1%                | 2.5% | 5%   | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |       |  |
|                   |      |      | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |       |  |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |  |
| 19.8              | 17.8 | 15.7 | DB                         | 52.6 | 102.8              | 2.6 | 1.6                                                | 50.7 | 103.9      | 49.2 | 104.9      | 47.7 | 105.8      | 45.8 | 106.9 |  |
|                   |      |      | WB                         | 51.1 | 82.2               | 2.4 | 2.1                                                | 49.3 | 83.7       | 47.9 | 84.9       | 46.5 | 86.1       | 44.7 | 87.6  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 76.0        | 81.7  | 82.8  | 79.8  | 76.9  | 73.0  | 71.1  | 70.9  | 71.9  | 73.4  | 74.9  | 76.2  | 79.5  | (6)  |
|                                                                        | DBStd       | 5.92        | 4.58  | 3.74  | 3.88  | 3.94  | 4.08  | 4.21  | 4.58  | 4.63  | 5.10  | 4.72  | 4.76  | 4.36  | (7)  |
|                                                                        | HDD50       | 0           | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | (8)  |
|                                                                        | HDD65       | 17          | 0     | 0     | 0     | 0     | 0     | 4     | 8     | 3     | 2     | 0     | 0     | 0     | (9)  |
|                                                                        | CDD50       | 9484        | 983   | 919   | 924   | 808   | 714   | 634   | 649   | 678   | 703   | 772   | 786   | 914   | (10) |
|                                                                        | CDD65       | 4026        | 518   | 499   | 459   | 358   | 249   | 189   | 192   | 216   | 254   | 307   | 336   | 449   | (11) |
|                                                                        | CDH74       | 30739       | 5193  | 5151  | 3814  | 2559  | 1360  | 946   | 1098  | 1309  | 1509  | 1807  | 2214  | 3779  | (12) |
|                                                                        | CDH80       | 11668       | 2289  | 2335  | 1357  | 821   | 364   | 239   | 319   | 462   | 583   | 652   | 772   | 1475  | (13) |
| (14) Wind                                                              | WSAvg       | 6.9         | 6.9   | 6.9   | 6.1   | 6.1   | 5.8   | 6.1   | 6.9   | 7.5   | 8.0   | 7.8   | 7.4   | 7.1   | (14) |
| (15) Precipitation                                                     | PrecAvg     | 55.7        | 9.7   | 6.3   | 7.1   | 3.4   | 2.2   | 1.3   | 1.1   | 1.2   | 2.8   | 4.2   | 7.1   | 9.1   | (15) |
|                                                                        | PrecMax     | 77.0        | 15.4  | 13.5  | 13.7  | 7.4   | 5.4   | 3.2   | 3.1   | 3.2   | 6.1   | 8.5   | 11.6  | 17.9  | (16) |
|                                                                        | PrecMin     | 29.6        | 3.7   | 1.1   | 3.3   | 0.9   | 0.8   | 0.2   | 0.1   | 0.2   | 0.5   | 1.8   | 3.3   | 4.6   | (17) |
|                                                                        | PrecStd     | 10.1        | 3.1   | 3.1   | 3.1   | 1.7   | 1.1   | 0.7   | 0.8   | 0.7   | 1.4   | 2.0   | 2.3   | 3.0   | (18) |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 99.0  | 100.0 | 96.7  | 93.5  | 91.1  | 89.6  | 90.0  | 93.5  | 96.7  | 96.9  | 96.8  | 98.6  | (19) |
|                                                                        |             | MCWB        | 77.2  | 77.1  | 78.6  | 78.6  | 76.1  | 72.9  | 70.0  | 70.8  | 72.3  | 75.0  | 76.5  | 77.3  | (20) |
|                                                                        | 2%          | DB          | 96.5  | 97.0  | 93.2  | 91.0  | 87.6  | 85.9  | 86.3  | 89.4  | 91.5  | 91.4  | 91.7  | 94.9  | (21) |
|                                                                        |             | MCWB        | 77.5  | 76.9  | 77.7  | 76.5  | 73.5  | 71.1  | 70.2  | 70.0  | 71.1  | 74.0  | 75.4  | 76.5  | (22) |
|                                                                        | 5%          | DB          | 93.4  | 94.7  | 89.8  | 87.7  | 84.2  | 82.6  | 84.0  | 85.6  | 86.4  | 86.4  | 87.8  | 91.4  | (23) |
|                                                                        |             | MCWB        | 77.5  | 76.8  | 77.0  | 75.8  | 72.3  | 69.9  | 69.5  | 69.1  | 70.9  | 72.9  | 74.2  | 77.0  | (24) |
|                                                                        | 10%         | DB          | 91.0  | 91.5  | 87.4  | 84.4  | 80.8  | 79.1  | 80.4  | 81.0  | 82.3  | 82.6  | 84.2  | 87.9  | (25) |
|                                                                        |             | MCWB        | 77.3  | 77.1  | 77.1  | 74.8  | 71.3  | 69.3  | 68.7  | 69.0  | 70.4  | 72.2  | 73.7  | 76.6  | (26) |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 81.9  | 83.0  | 81.1  | 80.6  | 77.7  | 75.4  | 75.0  | 75.9  | 77.3  | 78.9  | 78.9  | 81.8  | (27) |
|                                                                        |             | MCDB        | 93.0  | 95.4  | 91.0  | 90.6  | 87.1  | 85.8  | 84.7  | 86.8  | 91.0  | 91.7  | 90.3  | 91.9  | (28) |
|                                                                        | 2%          | WB          | 80.3  | 80.8  | 79.7  | 78.8  | 75.6  | 72.9  | 72.7  | 73.3  | 74.6  | 76.0  | 77.1  | 79.5  | (29) |
|                                                                        |             | MCDB        | 90.5  | 91.2  | 88.3  | 86.8  | 83.8  | 80.7  | 81.1  | 82.7  | 84.8  | 86.4  | 87.8  | 90.2  | (30) |
|                                                                        | 5%          | WB          | 79.3  | 79.6  | 78.5  | 77.0  | 74.0  | 71.8  | 71.1  | 71.7  | 72.8  | 74.4  | 75.6  | 77.9  | (31) |
|                                                                        |             | MCDB        | 89.0  | 89.3  | 86.4  | 84.0  | 80.8  | 78.9  | 78.9  | 80.2  | 80.7  | 82.7  | 84.9  | 87.6  | (32) |
|                                                                        | 10%         | WB          | 78.1  | 78.2  | 77.5  | 75.7  | 72.4  | 70.4  | 69.6  | 69.9  | 71.5  | 73.1  | 74.5  | 76.7  | (33) |
|                                                                        |             | MCDB        | 87.2  | 87.2  | 84.5  | 82.3  | 78.7  | 76.6  | 76.4  | 77.1  | 78.6  | 79.9  | 82.2  | 85.3  | (34) |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 16.2  | 17.4  | 14.3  | 15.5  | 16.1  | 16.4  | 17.1  | 17.5  | 16.2  | 15.2  | 14.1  | 15.0  | (35) |
|                                                                        |             | MCDBR       | 20.8  | 22.1  | 18.9  | 19.8  | 20.3  | 21.5  | 23.1  | 25.2  | 24.7  | 23.2  | 21.7  | 21.1  | (36) |
|                                                                        | 5% WB       | MCWBR       | 7.5   | 7.3   | 7.0   | 8.0   | 8.7   | 9.1   | 9.5   | 9.4   | 8.6   | 8.7   | 8.3   | 8.2   | (37) |
|                                                                        |             | MCWBR       | 17.9  | 18.5  | 16.4  | 17.1  | 17.6  | 17.8  | 19.3  | 20.6  | 19.4  | 19.2  | 18.1  | 17.7  | (38) |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.417 | 0.414 | 0.412 | 0.406 | 0.386 | 0.388 | 0.387 | 0.421 | 0.505 | 0.476 | 0.442 | 0.430 | (40) |
|                                                                        |             | taud        | 2.457 | 2.447 | 2.461 | 2.463 | 2.495 | 2.473 | 2.457 | 2.341 | 2.083 | 2.232 | 2.355 | 2.406 | (41) |
|                                                                        | Ebn at Noon |             | 295   | 291   | 283   | 271   | 264   | 256   | 260   | 262   | 252   | 271   | 285   | 291   | (42) |
|                                                                        |             | Edh at Noon | 38    | 38    | 36    | 33    | 30    | 29    | 30    | 37    | 51    | 47    | 42    | 40    | (43) |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1887  | 1964  | 1685  | 1484  | 1182  | 1101  | 1142  | 1358  | 1453  | 1630  | 1687  | 1858  | (44) |
|                                                                        | RadStd      |             | 271   | 200   | 142   | 108   | 66    | 84    | 98    | 106   | 188   | 190   | 133   | 161   | (45) |

## Historical Trends

|      |                        | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                        |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (46) | Station Only           | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (47) | Regional (3 neighbors) | N/A   | N/A     | -3.04  | N/A     | -2.52 | -2.99 | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page